

June 27, 2026

Dr. Mehmet A. Begen
Dr. Prashant Chintapalli
Management Science Area Group
Ivey Business School, Western University
London, ON, Canada

Dear Dr. Begen and Dr. Chintapalli,

I am writing to apply for the Postdoctoral Associate position in Management Science at Ivey. I am completing my PhD in Applied Mathematics at the Illinois Institute of Technology, where my research sits at the intersection of combinatorial optimization, network design, and data-driven decision making. I am especially interested in problems where mathematical modeling can improve service systems in healthcare and public service.

My doctoral work develops *FalCom*, a Markov chain Monte Carlo framework for sampling hierarchical service regions and capacitated facility locations at scale. FalCom's genericity and scalability extend to a range of operational problems in healthcare, logistics, and public service delivery. A management science perspective from Ivey would help translate the methodology into decision-ready tools for the projects described in the research statement. Additionally, I would like to pursue a stochastic and flow-based extension of FalCom at Ivey. This sits within the stochastic-OR tradition where your group has methodological depth to help implement.

The Chicago primary care project is at the stage where your management science expertise would help shape it into a realistic policy paper for federal authorities and a decision-support tool for healthcare operators. This is the methodology-to-policy translation of my work that requires clinical collaborations and management science expertise to apply for funding opportunities.

Western's broader health-services ecosystem, in particular Dr. Amit Garg's clinical leadership in the DONOR Network would also provide the clinical infrastructure for a national-scale equity study in Canadian kidney allocation with a strong grant opportunity. FalCom type Markov chains can be used to evaluate the current donor service area boundaries, to identify provinces systematically underserved across the distribution of feasible alternatives, and to explore alternative policies within the existing infrastructure.

Beyond the research directions described in the research statement, I bring experience in routing problems, convex reformulations of non-convex problems, metaheuristic design, and data analysis. I have also taken graduate coursework in machine learning, statistical learning, reinforcement learning, and neural networks. I would be excited to contribute to ongoing projects and integrate machine learning algorithms while also developing an independent research agenda.

As a summary, I am drawn to Ivey because of its emphasis on management science, and operations research as well as its focus on public service and healthcare applications. I would welcome the opportunity to discuss in details. Thank you for your consideration.

Sincerely,

Alaittin Kirtisoglu
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